Channa Dissanayaka

AI/ML Engineer

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Profile

Prospective graduate student with 2+ years of experience in building end-to-end AI solutions. Experienced in Agentic AI, NLP, data engineering, predictive modeling, computer vision, and scientific AI systems. Skilled in designing and deploying machine learning pipelines, building scalable data workflows using Apache Spark and Apache Airflow. Hands-on experience in translating advanced AI research into production-level code, optimizing deep learning architectures, and deploying cloud-native ML services.

Education

BSc(Hons) in Artificial Intelligence **2022 – 2026**
University of Moratuwa, Sri Lanka

Relevant Coursework: Machine Learning, Deep Learning, Artificial Neural Networks, NLP, Machine Vision, Big Data and Data Warehousing, Expert Systems, Multi-Agent Systems, Time Series Analysis and Forecasting, Computational Statistics.

Experience

Intern ML Engineer – Orysys **Feb 2025 – Aug 2025**

- Developed an RAG chatbot by modifying and integrating advanced reasoning algorithms to improve response quality and contextual understanding.
- Built and maintained scalable ML data pipelines to automate data extraction schedules for large-scale financial datasets. Optimized data engineering workflows using Apache Spark, Apache Hadoop, and Apache Airflow to improve the efficiency and reliability of data processing pipelines..

Freelancing AI/ML Developer – Fiverr **Jan 2023 – Present**

- Delivered end-to-end AI/ML solutions for international and local clients, specializing in computer vision, predictive modeling, and custom machine learning applications.

Technical Skills

Deep Learning: TensorFlow, Keras, PyTorch
Computer Vision: YOLO, OpenCV, Roboflow
LLMs and NLP: RAG, Transformers, LangChain, HuggingFace, Vector Databases
Data Engineering: Airflow, Hadoop, Spark
Data Science & ML Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, SciPy, JAX
Programming Languages: Python, Java, C, R, Prolog
Databases: MSSQL, MySQL
Cloud: GCP, Runpod
Other: Git, Docker, Firebase

Publications

The Evolution of the AlphaFold Architecture (Preprint) Link to paper

- A review paper analyzing architectural advancements in protein structure prediction.

Projects

AlphaFold-Based Protein Conformational Ensemble Prediction – Final Year Research GitHub Link
Designed and developed a training-free, inference-time pipeline for AlphaFold-based protein conformational prediction using PyTorch, JAX, and ColabFold. Implemented a dual-stochastic framework integrating the DBSCAN clustering algorithm, localized random column masking, and active inference-time dropout. Evaluated the pipeline against the standardized OC23 dataset, achieving a 78.3% success rate while delivering a 10X reduction in computational cost by generating only 100 models per target.

Multi Agent Customer Service Chatbot (Ongoing) GitHub Link

Building a full-stack AI customer service chatbot for a coffee shop that takes orders and answers menu queries, filters out irrelevant conversations, and recommends products using a Market Basket Analysis engine. The system combines prompt engineering, Retrieval-Augmented Generation (RAG), and modular agent-based architecture to route and ground responses, backed by a deployed LLM and vector search, with a React Native mobile front end.

Key contributions:

- Deployed a Llama-based LLM and embedding models on RunPod GPU infrastructure.
- Implemented Retrieval-Augmented Generation (RAG) with a Pinecone vector database to ground responses in menu and business data.
- Built a modular agent-based system to separate concerns such as order-taking, FAQ handling, and conversation filtering.
- Developed a Market Basket Analysis recommendation engine to suggest products.
- Set up Firebase as the backend database for orders, menu, and app data.
- Built the end-to-end React Native mobile app, integrating the chat interface with Firebase and RunPod endpoints.

AI Assisted Carbon Footprint Monitoring System GitHub Link

Developed an AI-powered carbon footprint monitoring platform for food supply chains that estimates CO_2e emissions from farm to market. Built machine learning models for emission prediction, developed the frontend and backend, and deployed ML models on Google Cloud Platform.

Key contributions:

- Developed emission prediction models.
- Built REST APIs using Flask.
- Developed the frontend using React.
- Developed backend using Spring Boot.
- Deployed ML models on GCP.

Football Analysis System GitHub Link

Developed an AI-powered football analytics system using YOLOv8 for player detection and tracking on a custom dataset. Implemented K-Means clustering for automated team classification, optical flow for camera motion compensation, and OpenCV perspective transformation to estimate player speed, distance covered, and positional movement from broadcast footage.